

Carlos Henrique Caloete Pena

Lead Data Engineer

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📍 Recife, Brazil · Brazilian/Portuguese citizenship (EU work authorization)

Summary

Lead Data Engineer with 8+ years and an M.Sc. in Computer Science (UFPE, GPA 4.0/4.0). Currently leading a multi-squad R&D group that spans both sides of the data/AI boundary: dimensional warehouses and event-driven services alongside computer vision and LLM systems, all running in production at retail scale.

Experience

Ferreira Costa	Top 5 Retail/E-commerce in Brazil	Recife, Brazil · Jul 2021 - Present
Lead Data Engineer		Apr 2025 - Present
◦ Lead 6 cross-functional agile squads (~17 data engineers, data scientists, and backend developers) across data platform, computer vision, search, and generative AI, owning delivery alignment with Product Owners plus hiring, performance reviews, and career development. ◦ Migrated enterprise analytics off a legacy Oracle system into BigQuery, decoupling reporting from the operational database. ◦ Modeled the warehouse as Kimball star schemas across Bronze/Silver/Gold layers: 30+ models spanning conformed dimensions (branch, category, product, customer, date) over sales, inventory, and pricing facts, orchestrated with SQLMesh and Dagster with continuous testing, schema contracts, and lineage. Now extending into operational workflows, starting with automated price calculation (in development). ◦ Shipped semantic search for long-tail queries using AWS SageMaker-hosted embeddings, increasing CTR by +3.4 pp (+9.6% relative) with no material impact on response latency. ◦ Built and scaled the product catalog platform from initial design (MongoDB, RabbitMQ) to an event-driven architecture (FastAPI, FastStream) centralizing inventory, pricing, and media, with Google Merchant Center integration; load-tested (k6) to 55 req/s per pod. ◦ Delivered a secure FastAPI/FastMCP platform exposing enterprise data (Oracle, GCP, internal APIs) to LLM agents via MCP interfaces, with authentication, rate limiting, and governance controls, giving ~36 internal users self-service access without direct database exposure.		
Data Engineer Specialist		Jun 2024 - Mar 2025
◦ Led four agile squads (12-15 data engineers, data scientists, and backend developers) in R&D of computer vision and generative AI solutions, translating high-level business objectives into technical roadmaps with Product Owners. ◦ Architected an image processing pipeline combining generative AI, deep learning, and traditional computer vision on AWS SageMaker (Python, OpenCV, FastAPI) for hands-off product media enhancement and quality control. ◦ Built chatbots and workflows using Generative AI with RAG, orchestrated via n8n.		
Senior Data Engineer		Jul 2021 - May 2024
◦ Engineered a 24/7 Shipping System capable of handling over 13K concurrent users during peak events like Black Friday, with integrated observability and a stack of Python, Oracle, PostgreSQL, Redis, and AWS (ECR, EKS, S3). ◦ Designed and maintained query acceleration layers (materialized views, scheduled jobs, Redis caches) and workflow orchestration with Dagster. ◦ Oversaw CI/CD for 19 production deployments (APIs, workers, and DAGs) using GitLab CI/CD, AWS, and Rancher. ◦ Built a custom product search engine using Elasticsearch, ranked on Google Analytics behavioral signals, reaching 41.8% search-page CTR on head queries and 31.4% on tail queries.		
NCR (formerly OKI Brasil)	Banking automation and self-service technology	Recife, Brazil
Data Scientist		Jun 2018 - Jun 2021

- **Designed** image quality measures for capturing ID photos (facial close-up) in compliance with ICAO standards.
- **Researched and prototyped** computer vision and deep learning solutions.
- **Presented** weekly progress to administrative and technical stakeholders at NCR São Paulo, leading discussions on emerging AI techniques and defining next steps from new requirements and system limitations.

RobôCIn - UFPE	Student robotics research group	Recife, Brazil
Co-founder		2015 - 2021

- **Led** research teams applying deep learning and reinforcement learning to autonomous robot soccer at UFPE's Centro de Informática (see [P2](#)).
- **Engineered** vision and control systems (OpenCV, ROS, Qt), including a human-size domestic robot for the RoboCup@Home category.

Education

Universidade Federal de Pernambuco (UFPE)	Recife, Brazil
Master of Science in Computer Science - GPA 4.0/4.0	Graduated Oct 2022
◦ Thesis: Segmentation of medical images (Advisor: Tsang Ing Ren); see P1	

Universidade Federal de Pernambuco (UFPE)	Recife, Brazil
Bachelor of Science in Computer Engineering - GPA 8.67/10 (2nd in class)	Graduated Dec 2019

Publications

- P1 [An Ensemble Learning Method for Segmentation Fusion](#)**
Carlos H. C. Pena, Tsang Ing Ren, Pedro D. Marrero Fernandez, Fidel A. Guerrero-Peña, Alexandre Cunha
International Joint Conference on Neural Networks (IJCNN), 2022
- P2 [An Analysis of Reinforcement Learning Applied to Coach Task in IEEE Very Small Size Soccer](#)**
Carlos H. C. Pena, M. G. Machado, M. S. Barros, J. D. P. Silva, L. D. Maciel, T. Ing Ren, E. N. S. Barros, P. H. M. Braga, H. F. Bassani
Latin American Robotics Symposium (LARS), 2020

Skills

Programming Languages: Python, SQL, PL/SQL, C/C++

AI and Machine Learning: Computer Vision, Deep Learning, Machine Learning, PyTorch, Generative AI, RAG, OpenCV, MCP

Data and Cloud: AWS (EKS, SageMaker, ECR, S3), GCP (BigQuery), Oracle, PostgreSQL, Elasticsearch, MongoDB, Redis, RabbitMQ, Dagster (Airflow equivalent), SQLMesh (dbt equivalent), Pandas, Polars

Tools and Practices: Git, GitLab CI/CD, Docker, Kubernetes, Rancher, FastAPI, FastStream, n8n, k6, Agile/Scrum

Leadership: Hiring and mentoring, performance management, stakeholder alignment, cross-functional delivery

Spoken Languages: Portuguese (native), English (full professional proficiency)

Awards and Honors

IT Professional of the Year , FerreiraCosta/FCxLabs	2023
1st Place Team , Microsoft College Code Competition UFPE	2019
Latin American IEEE Very Small Size Soccer , top-5 finishes across multiple years	2017, 2020

Certifications

- Data Engineering Specialization (DeepLearning.AI / AWS, 2025): Data Storage and Queries; Data Modeling, Transformation, and Serving; Source Systems, Data Ingestion, and Pipelines; Introduction to Data Engineering (coursework includes Apache Spark, AWS Glue)
- Claude Certified Architect - Foundations (Anthropic, 2026); Kafka (Alura, 2023); Retrieval Augmented Generation (DeepLearning.AI, 2025); LlamaIndex: creating a chatbot with the RAG technique (Alura, 2024)
- Agile Management Practices SC-AMP (Agile Institute Brazil, 2022); Scrum (Alura, 2022)